

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains correct vector	AO1.1b	B1	$\begin{pmatrix} -4 \\ -3 \\ 6 \end{pmatrix}$
(b)	Obtains one other edge as vector	AO1.1a	M1	$\vec{BC} = \begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix}$
	Obtains $\vec{DC}$ correctly Or Obtains correctly both $\vec{BC}$ and $\vec{AD}$ Or Obtains correctly both $\vec{CB}$ and $\vec{DA}$	AO1.1b	A1	$\vec{AD} = \begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix}$ $\vec{DC} = \begin{pmatrix} -4 \\ -3 \\ 6 \end{pmatrix}$
	Obtains length of one edge (or its square)	AO1.1a	M1	$AB = \sqrt{(-4)^2 + (-3)^2 + 6^2}$ $= \sqrt{61}$
	Obtains two correct lengths of different edges	AO1.1b	A1	$AD = \sqrt{1^2 + 5^2 + (-1)^2}$ $= 3\sqrt{3}$
	Completes rigorous argument to show $ABCD$ is a parallelogram and not a rhombus	AO2.1	R1	$\vec{AB} = \vec{DC}$ $ABCD$ must be a parallelogram $AB \neq AD$ $ABCD$ is not a rhombus
				Total 6 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Circles correct answer	AO1.1b	B1	29
(b)	Circles correct answer	AO2.2a	B1	$90^\circ < \theta < 135^\circ$
				Total 2 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Sums the forces given correctly	AO1.1b	B1	$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$ $= 33\mathbf{i} - 11\mathbf{j}$
	Uses Pythagoras to find the magnitude of the vector and obtains correct magnitude (given to 3 sig figs)	AO1.1b	B1	$ \mathbf{F} = \sqrt{33^2 + (-11)^2}$ $= 34.8 \text{ N (3 sf)}$
(ii)	Uses trig expression with appropriate values	AO1.1a	M1	$\tan \theta = \frac{11}{33}$
	Obtains correct angle (given to nearest 0.1°)	AO1.1b	A1	$\theta = \tan^{-1}\left(\frac{1}{3}\right)$ $= 18.4^\circ \text{ (3 sf)}$ OR $\sin \theta = \frac{11}{\sqrt{1210}}$ $\theta = \sin^{-1}\left(\frac{11}{\sqrt{1210}}\right)$ $= 18.4^\circ \text{ (3 sf)}$ OR $\cos \theta = \frac{33}{\sqrt{1210}}$ $\theta = \cos^{-1}\left(\frac{33}{\sqrt{1210}}\right)$ $= 18.4^\circ \text{ (3 sf)}$
(b)	States negative of 'their' part (a)(i)	AO2.2a	B1F	$\mathbf{F}_4 = -33\mathbf{i} + 11\mathbf{j}$
				Total 5 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains correct horizontal component of the initial velocity	AO1.1b	B1	$2.5U = 40$ $U = 16$

	Forms equation to find vertical component of initial velocity	AO3.3	M1	$-10 = 2.5V - 0.5 \times 9.81 \times 2.5^2$
	Obtains correct vertical component of initial velocity	AO1.1b	A1	$V = 8.2625$
	Forms equation for vertical component of velocity at height 3 using 'their' derived values for U and V	AO3.4	M1	$v_y^2 = 8.2625^2 + 2 \times (-9.81) \times 3$
	Obtains correct component of velocity	AO1.1b	A1	$v_y = 3.067...$
	Correct final speed with units, correct for 'their' U and v_y FT applies only if both M1 marks have been awarded	AO3.2a	A1F	$v = \sqrt{16^2 + 3.067^2} = 16.3 \text{ ms}^{-1}$
(b)	States 'their' value of horizontal component of the initial velocity from part (a)	AO3.4	A1F	16 m s^{-1}
(c)	Explains that horizontal velocity has been assumed to be constant in their model and that this is not likely to be true, with valid reasoning	AO3.5b	E1	It was assumed that there were no resistance forces acting on the ball which is unlikely to be true in reality. The horizontal speed of the ball is likely to vary... air resistance would slow the ball down, wind might speed the ball up
				Total 8 marks

Q5.

(a) $(F =) 9\mathbf{i} - 3\mathbf{j} + 5\mathbf{i} + 8\mathbf{j} - 7\mathbf{i} + 3\mathbf{j} = 7\mathbf{i} + 8\mathbf{j}$

M1: Adding the three forces with one component correct.

A1: Correct answer.

M1A1

2

(b) $(F =) \sqrt{7^2 + 8^2} = \sqrt{113} = 10.6 \text{ N}$

M1: Finding magnitude with a + sign.

A1F: Correct magnitude. Accept AWRT 10.63 and $\sqrt{113}$

Follow through incorrect answers to part (a).

M1A1F

2

(c) $(a =) \frac{\sqrt{113}}{5} = 2.13 \text{ m s}^{-2}$

M1: Dividing their force from part (a) or magnitude by 5.

A1F: Correct acceleration.

Accept 2.12 (from truncation or $10.6 / 5$) or $\frac{\sqrt{113}}{5}$ or AWRT 2.13.

Follow through incorrect answers to parts (a) and (b).

Seeing just $\mathbf{a} = 1.4\mathbf{i} + 1.6\mathbf{j}$ scores M1 A0

M1A1F

2

(d) $\cos \alpha = \frac{7}{\sqrt{113}} \text{ or } \frac{7}{10.6}$

M1: Trig equation to find the angle with: cos with 7 or 8 in the numerator and $\sqrt{113}$ in denominator

OR

$\sin \alpha = \frac{8}{\sqrt{113}} \text{ or } \frac{8}{10.6}$

sin with 7 or 8 in the numerator and $\sqrt{113}$ in denominator

OR

$\tan \alpha = \frac{8}{7}$

tan with 7 and 8 in any position

A1F: Correct equation.

M1A1F

$(\alpha =) 48.8^\circ$

A1F: Correct angle. Accept 49° or AWRT 49°

Follow through incorrect answers to parts (a) and (b).

A1F

3

[9]

Q6.

(a) $v = \int a \, dt$

$$= (20t^2 + t^3)\mathbf{i} - 5e^{-4t}\mathbf{j} + \mathbf{c}$$

*M1 for either term correct.
Condone no '+ c'.*

M1A1

When $t = 1$,

$$6\mathbf{i} - 5e^{-4}\mathbf{j} = 21\mathbf{i} - 5e^{-4}\mathbf{j} + \mathbf{c}$$

Finding '+ c'; not using $\mathbf{c} = 6\mathbf{i} - 5e^{-4}\mathbf{j}$.

M1

$$\mathbf{c} = -15\mathbf{i}$$

A1

$$\mathbf{v} = (20t^2 + t^3 - 15)\mathbf{i} - 5e^{-4t}\mathbf{j}$$

A1

5

(b) When $t = 0$, $\mathbf{v} = -15\mathbf{i} - 5\mathbf{j}$

M1

Speed is $\sqrt{15^2 + 5^2}$

M1

$$= 15.8 \text{ m s}^{-1}$$

Accept $5\sqrt{10}$

A1

3

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

[8]

Q7.

(a) using $\mathbf{F} = m\mathbf{a}$:

ie dividing by 50

M1

$$\mathbf{a} = (6t - 1.2t^2)\mathbf{i} + 2e^{-2t}\mathbf{j}$$

A1

2

(b) $\mathbf{v} = \int \mathbf{a} \, dt$

$$= (3t^2 - 0.4t^3)\mathbf{i} - e^{-2t}\mathbf{j} + \mathbf{c}$$

when $t = 0$, $\mathbf{r} = 7\mathbf{i} - 4\mathbf{j}$
condone lack of + c; M1 one term correct

M1A1

$\mathbf{c} = 7\mathbf{i} - 3\mathbf{j}$
 $\mathbf{v} = (7 + 3t^2 - 0.4t^3)\mathbf{i} - (3 + e^{-2t})\mathbf{j}$
ft from ke^{-2t} in (b); just adding $7\mathbf{i} - 4\mathbf{j}$, m0 accept unsimplified. CAO

m1A1

4

(c) when $t = 1$, $\mathbf{v} = 9.6\mathbf{i} - 3.135\mathbf{j}$
ft from (b)

M1A1

$$\text{speed} = \sqrt{9.6^2 + 3.135^2}$$

m1

$$= 10.1 \text{ ms}^{-1}$$

ft from (b)

A1

4

[10]

Q8.

(a) $\mathbf{r} = (-\mathbf{i} + 3\mathbf{j})t + \frac{1}{2}(0.1\mathbf{i} - 0.2\mathbf{j})t^2$
M1: Using constant acceleration equation to get $\mathbf{r}$.
A1: Correct expression for $\mathbf{r}$. Allow equivalent column vector answer.

M1A1

2

(b) $3t - 0.1t^2 = 0$

$$t(3 - 0.1t) = 0$$

$$t = 0 \text{ or } t = 30$$

M1: Putting their $\mathbf{j}$ component equal to zero to form a quadratic equation.

A1: Correct equation.

M1A1

$t = 30$ seconds

A1: For 30 seconds. No need to see $t = 0$.

A1

3

(c) $\mathbf{v} = (0.1t - 1)\mathbf{i} + (3 - 0.2t)\mathbf{j}$

*B1: Correct expression for the velocity in terms of t .
Can be implied by subsequent working in terms of t .*

B1

$$0.1t - 1 = -(3 - 0.2t)$$

$$2 = 0.1t$$

M1: For $0.1t - 1 = \pm(3 - 0.2t)$. May be with their components if velocity stated incorrectly.

A1: Correct equation.

M1A1

$$t = 20$$

A1: $t = 20$

A1

$$\mathbf{v} = \mathbf{i} - \mathbf{j}$$

$$v = \sqrt{2} = 1.41 \text{ ms}^{-1}$$

dM1: finding velocity and speed at their time

dM1

A1: Correct speed.

A1

Special cases

If the equation in t in line 2 is not seen:

then seeing $t = 20$ and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ and $v = 1.41$

award 4 out of 6

or

then seeing $t = 20$ and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ award 2 out of 6

6

[11]

Q9.

(a) $\mathbf{r} = (-\mathbf{i} + 3\mathbf{j})t + \frac{1}{2}(0.1\mathbf{i} - 0.2\mathbf{j})t^2$

M1: Using constant acceleration equation to get r .

A1: Correct expression for r . Allow equivalent column vector answer.

M1A1

2

(b) $3t - 0.1t^2 = 0$

$t(3 - 0.1t) = 0$

$t = 0$ or $t = 30$

M1: Putting their j component equal to zero to form a quadratic equation.

A1: Correct equation.

M1A1

$t = 30$ seconds

A1: For 30 seconds. No need to see $t = 0$.

A1

3

(c) $\mathbf{v} = (0.1t - 1)\mathbf{i} + (3 - 0.2t)\mathbf{j}$

*B1: Correct expression for the velocity in terms of t .
Can be implied by subsequent working in terms of t .*

B1

$0.1t - 1 = -(3 - 0.2t)$

$2 = 0.1t$

M1: For $0.1t - 1 = \pm(3 - 0.2t)$. May be with their components if velocity stated incorrectly.

A1: Correct equation.

M1A1

$t = 20$

A1: $t = 20$

A1

$\mathbf{v} = \mathbf{i} - \mathbf{j}$

$v = \sqrt{2} = 1.41 \text{ ms}^{-1}$

dM1: finding velocity and speed at their time

dM1

A1: Correct speed.

A1

Special cases

If the equation in t in line 2 is not seen:

then seeing $t = 20$ and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ and $v = 1.41$

award 4 out of 6

or

then seeing $t = 20$ and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ award 2 out of 6

6

[11]

Q10.

(a) $\mathbf{v} = (4\mathbf{i} + 0.5\mathbf{j}) + (-0.4\mathbf{i} + 0.2\mathbf{j})t$

M1: Use of constant acceleration equation to find $\mathbf{v}$
with $\mathbf{u} \neq 0\mathbf{i} + 0\mathbf{j}$

A1: Correct $\mathbf{v}$.

(Could be done as a column vector.)

M1A1

2

(b) (i) $\mathbf{v} = (4\mathbf{i} + 0.5\mathbf{j}) + (-0.4\mathbf{i} + 0.2\mathbf{j}) \times 22.5$

M1: Substitution of 22.5 into their expression for the
velocity, even if no marks awarded in part (a).

M1

$= -5\mathbf{i} + 5\mathbf{j}$

A1: Correct velocity CAO

(Could be done using column vectors.)

A1

2

(ii) North-west

B1: Correct statement of direction. Accept 315° .
Must follow from correct answer to (b)(i).

B1

1

(c) $(\mathbf{v}) = (4 - 0.4t)\mathbf{i} + (0.5 + 0.2t)\mathbf{j}$

B1: Grouping $\mathbf{i}$ and $\mathbf{j}$ components at some point in the
solution.

(Could be done using column vectors.)

Allow $5 = (4 - 0.4t)\mathbf{i} + (0.5 + 0.2t)\mathbf{j}$

B1

$5^2 = (4 - 0.4t)^2 + (0.5 + 0.2t)^2$

M1: Seeing both components of their velocity squared
and added

A1: Correct equation. (Condone including $\mathbf{i}$ and $\mathbf{j}$.)

For example: $5 = (4 - 0.4t)i^2 + (0.5 + 0.2t)j^2$ scores B1M1A0
 $5^2 = (4 - 0.4t)i^2 + (0.5 + 0.2t)j^2$ scores B1M1A1

M1A1

$$0.2t^2 - 3t - 8.75 = 0$$

A1: Any correct simplified quadratic equation,
 with exactly three terms.

A1

$$t^2 - 15t - 43.75 = 0$$

$$t = 17.5 \text{ or } t = -2.5$$

dM1: Solving the quadratic equation.
 (Allow one substitution error in correctly quoted formula)
 Candidates with an incorrect quadratic equation must
 show method to get dM1.

dM1

$$t = 17.5$$

A1: Correct positive solution stated.

A1

6

[11]

Q11.

(a) $\mathbf{r} = \int \mathbf{v} \, dt = (4t + t^3)\mathbf{i} + (12t - 4t^2)\mathbf{j} + \mathbf{c}$

M1: either i or j term correct.
 Condone no $\mathbf{c}$

M1A1

When $t = 0$, $\mathbf{r} = 5\mathbf{i} - 7\mathbf{j}$
 $\mathbf{c} = 5\mathbf{i} - 7\mathbf{j}$

Any attempt at $\mathbf{c}$

M1

$$\mathbf{r} = (5 + 4t + t^3)\mathbf{i} + (-7 + 12t - 4t^2)\mathbf{j}$$

A1

4

(b) $\mathbf{a} = \frac{d\mathbf{v}}{dt}$

$$\mathbf{a} = 6t \mathbf{i} - 8 \mathbf{j}$$

M1: either term correct

M1A1

2

(c) Using $\mathbf{F} = m\mathbf{a}$

Or: using $\mathbf{F} = m\mathbf{a}$

M1

$$\mathbf{F} = 2(6t \mathbf{i} - 8 \mathbf{j})$$

$$\mathbf{F} = 2(6t \mathbf{i} - 8 \mathbf{j})$$

$$= 12t \mathbf{i} - 16 \mathbf{j}$$

$$\text{When } t = 1, \mathbf{F} = 12 \mathbf{i} - 16 \mathbf{j}$$

A1

$$\therefore \text{Magnitude of force is } (144t^2 + 256)^{\frac{1}{2}} \text{ when } t = 1$$

$$\text{Magnitude of force is } (144 + 256)^{\frac{1}{2}}$$

M1

$$= 20 \text{ N}$$

$$= 20 \text{ N}$$

A1

4

[10]

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Q12.

(a) $\mathbf{v} = (0.5\mathbf{i} + 0.375\mathbf{j}) \times 20 (=10\mathbf{i} + 7.5\mathbf{j})$

M1: Calculating velocity with $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$ and $t = 20$.

A1: Correct expression for velocity.

M1A1

$$v = \sqrt{10^2 + 7.5^2} = 12.5 \text{ ms}^{-1}$$

dM1: Calculating speed.

A1: Correct speed.

dM1A1

4

(b) $\tan \theta = \frac{0.5}{0.375} \text{ or } \frac{10}{7.5} \left(\text{or } \tan \theta = \frac{0.375}{0.5} \text{ or } \frac{7.5}{10} \right)$

M1: Using trig to find angle. Can be inverted as shown in brackets.

A1F: Correct trig expression with any correct equivalent fraction.

M1A1F

$$\theta = 053^\circ$$

A1: Correct angle to the nearest degree. Accept 53°.

A1

OR

$$\cos\theta = \frac{7.5}{12.5} \text{ or } \frac{0.375}{0.625} \left(\text{or } \cos\theta = \frac{10}{12.5} \right)$$

(M1A1F)

$$\theta = 053^\circ$$

Note: For 37° award M1A0A0

But for 90 – 37 = 53° award M1A1A1.

For 127°, award M1A1A0

(A1)

OR

$$\sin\theta = \frac{10}{12.5} \text{ or } \frac{0.5}{0.625} \left(\text{or } \sin\theta = \frac{7.5}{12.5} \right)$$

Note: 53.1° as final answer scores M1A1A0

(M1A1F)

$$\theta = 053^\circ$$

Condone finding angle from acceleration or position vector.

(A1)

©2025 Exam Papers Practice. All rights reserved.

3

(c) $(\mathbf{r}) = \frac{1}{2} (0.5\mathbf{i} + 0.375\mathbf{j}) t^2 (= 0.25 t^2 \mathbf{i} + 0.1875 t^2 \mathbf{j})$

M1: Finding an expression for position vector in terms of t.

A1: Correct position vector.

M1A1

$$500^2 = (0.25 t^2)^2 + (0.1875 t^2)^2$$

dM1: Using distance to form an equation for t.

A1: Correct equation.

dM1A1

$$t = 4\sqrt{\frac{500^2}{0.25^2 + 0.1875^2}} = 40 \text{ seconds}$$

A1: Correct time.

A1

OR

$$a = 0.625$$

M1: Finding magnitude of acceleration.

A1: Correct acceleration

(M1A1)

$$500 = \frac{1}{2} 0.625 t^2$$

dM1: Using distance to form an equation for t .

A1: Correct equation.

(dM1A1)

$$t = 40$$

A1: Correct time.

(A1)

OR

$$400 = \frac{1}{2} 0.5 t^2 \text{ or } 300 = \frac{1}{2} 0.375 t^2$$

M1: Working with one component.

A1: Correct distance (300 or 400)

A1: Correct equation.

(M1A1)

(A1)

$$t^2 = 1600$$

dM1: Solving for t .

(dM1)

$$t = 40$$

A1: Correct t .

Note: $500 \div 12.5 = 40$ is not acceptable and scores 0

(A1)

5

[12]

Q13.

(a) $\mathbf{v} = 4.5\mathbf{i} + 2\mathbf{j}$

B1: Correct velocity

B1
1

(b) $\mathbf{v} = 4.5\mathbf{j}$

B1: Correct velocity

B1
1

(c) Due North

B1: Correct direction

B1
1

(d) $4.5\mathbf{j} = 4.5\mathbf{i} + 2\mathbf{j} + 5\mathbf{a}$

M1: Use of a constant acceleration equation to find a

A1: Correct equation

M1A1

$$\mathbf{a} = \frac{-4.5\mathbf{i} + 2.5\mathbf{j}}{5} = -0.9\mathbf{i} + 0.5\mathbf{j}$$

A1: Correct acceleration

A1
3

(e) $\mathbf{F} = 250(-0.9\mathbf{i} + 0.5\mathbf{j}) = -225\mathbf{i} + 125\mathbf{j}$

M1: Using Newton's second law.

M1

$$F = \sqrt{225^2 + 125^2} = 257 \text{ N}$$

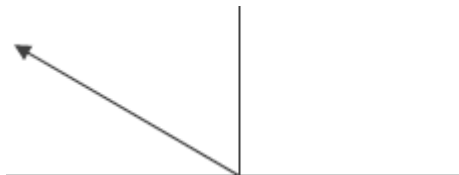
M1: Finding magnitude.

A1: Correct magnitude of the force

M1A1

3

(f)



B1: Diagram with arrow in the correct quadrant.

B1

$$\tan^{-1}\left(\frac{125}{225}\right) = 29.1^\circ$$

M1: Use of trig to find an angle.

M1

A1: Correct angle from working.

A1

OR

$$\tan^{-1}\left(\frac{0.5}{0.9}\right) = 29.1^\circ$$

$$180 - 29.1 = 151^\circ$$

A1: Finding required angle correctly.

A1

4

[13]

Q14.

(a) $(8\mathbf{i} + 12\mathbf{j}) + (4\mathbf{i} - 4\mathbf{j}) = 12\mathbf{i} + 8\mathbf{j}$

M1: Adding forces to find resultant.

A1: Correct resultant force.

M1A1

2

(b) $4\mathbf{a} = 12\mathbf{i} + 8\mathbf{j}$ or $(\mathbf{a} =) \frac{12\mathbf{i} + 8\mathbf{j}}{4}$

M1: Use of Newton's second law with $4\mathbf{a}$ and their answer to part (a).

M1

$(\mathbf{a} =) 3\mathbf{i} + 2\mathbf{j}$ **AG**

A1: Correct acceleration from correct equation.

A1

2

(c) (i) $40\mathbf{i} + 32\mathbf{j} = \mathbf{v} + (3\mathbf{i} + 2\mathbf{j}) \times 20$
 $40\mathbf{i} + 32\mathbf{j} = \mathbf{v} + 60\mathbf{i} + 40\mathbf{j}$

B1: Seeing $60\mathbf{i} + 40\mathbf{j}$ or $(3\mathbf{i} + 2\mathbf{j}) \times 20$

B1

M1: Use of constant acceleration equation with $t = 20$ and $\mathbf{a} = 3\mathbf{i} + 2\mathbf{j}$.

M1

$$\mathbf{v} = (40\mathbf{i} + 32\mathbf{j}) - (60\mathbf{i} + 40\mathbf{j})$$

AG

$$= -20\mathbf{i} - 8\mathbf{j}$$

*A1: Correct velocity from correct working, with one of the intermediate lines of working (or equivalent) shown.
 Note: Candidates may use $\mathbf{u}$ instead of $\mathbf{v}$ in their working.*

Example:

Starting with

$$\mathbf{v} = 40\mathbf{i} + 32\mathbf{j} + (3\mathbf{i} + 2\mathbf{j}) \times 20$$

Scores B1M1A0.

A1

3

Note on Verification Method:

$$\mathbf{v} = (-20\mathbf{i} - 8\mathbf{j}) + (3\mathbf{i} + 2\mathbf{j}) \times 20 \quad \text{B1M1}$$

$$= (-20 + 60)\mathbf{i} + (-8 + 40)\mathbf{j}$$

$$= 40\mathbf{i} + 32\mathbf{j} \quad \text{A1}$$

Similarly, verification to confirm acceleration from the two velocities is acceptable.

(ii) $(\mathbf{v} =) (-20\mathbf{i} - 8\mathbf{j}) + (3\mathbf{i} + 2\mathbf{j})$

B1: Correct velocity vector.

Note " $\mathbf{v} =$ " does not need to be seen.

B1

1

(iii) $(\mathbf{v} =) (3t - 20)\mathbf{i} + (2t - 8)\mathbf{j}$

M1: Velocity vector seen split into components. Condone omission of $\mathbf{i}$ and $\mathbf{j}$

Note: This can be implied by later working, such as the second line of this solution.

M1

$$(3t - 20)^2 + (2t - 8)^2 = 8^2$$

dM1: Equation based on speed of 8.

dM1

A1: Correct unsimplified equation.

A1

$$13t^2 - 152t + 400 = 0$$

A1: Simplified quadratic equation

A1

$$t = \frac{152 \pm \sqrt{152^2 - 4 \times 13 \times 400}}{2 \times 13}$$

dM1: Solving quadratic equation, to obtain two solutions.

dM1

$$t = 4 \text{ or } t = 7.69$$

A1: Both correct solutions. Accept

AWRT 7.7 or 7.6 or $\frac{100}{13}$.

Note: Using calculator to solve quadratic is acceptable.

A1

6

[14]

Q15.

(a) $\mathbf{v} = 9\mathbf{i} + 7\mathbf{j}$

B1: Correct initial velocity.

B1

1

(b) $\mathbf{v} = (9 - 0.01t)\mathbf{i} + (7 - 0.03t)\mathbf{j}$

M1: Writing $\mathbf{v}$ in the form $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

$= (9\mathbf{i} + 7\mathbf{j}) + (-0.01\mathbf{i} - 0.03\mathbf{j})t$
 $\mathbf{a} = -0.01\mathbf{i} - 0.03\mathbf{j}$

A1: Correct acceleration.

A1

2

(c) $9 - 0.01t = -(7 - 0.03t)$

M1: Equation involving both $\mathbf{i}$ and $\mathbf{j}$ components of their velocity.

A1: Correct equation, from their acceleration.

M1A1

$$t = \frac{16}{0.04} = 400$$

A1: Correct time, from their acceleration.

A1

3

[6]

Q16.

(a) $10\mathbf{a} = 9\mathbf{i} + 12\mathbf{j}$

M1: Application of Newton's second Law with $m = 10$ in vector form.

M1

$\mathbf{a} = (0.9\mathbf{i} + 1.2\mathbf{j}) \text{ m s}^{-2}$

*A1: Correct acceleration.
If acceleration incorrect follow their value through for the rest of this question.*

A1

2

(b) (i) $\mathbf{r}(5) = (2.2\mathbf{i} + 1\mathbf{j}) \times 5 + \frac{1}{2} (0.9\mathbf{i} + 1.2\mathbf{j}) \times 5^2$

M1: Use of constant acceleration to find position vector at $t = 5$, with $\mathbf{u} \neq 0\mathbf{i} + 0\mathbf{j}$.

M1

$= 22.25\mathbf{i} + 20\mathbf{j}$

A1F: Correct position vector, for candidate's acceleration which must be a vector. Allow $22.3\mathbf{i} + 20\mathbf{j}$.

A1F

$d = \sqrt{22.25^2 + 20^2} = 29.9 \text{ metres}$

dM1: Calculation of distance from position vector. Must see + sign.

dM1

*A1F: Correct distance, for their acceleration.
Accept 30 from $22.3\mathbf{i} + 20\mathbf{j}$.*

A1F

4

(ii) $\mathbf{v} = (2.2\mathbf{i} + 1\mathbf{j}) + (0.9\mathbf{i} + 1.2\mathbf{j})t$

M1: Use of constant acceleration equation to find an expression for $\mathbf{v}$, with $\mathbf{u} \neq 0\mathbf{i} + 0\mathbf{j}$

M1

A1F: Correct $\mathbf{v}$ for their acceleration.

A1F

2

(iii) $\mathbf{v} = (2.2 + 0.9t)\mathbf{i} + (1 + 1.2t)\mathbf{j}$
 $2.2 + 0.9t = 1 + 1.2t$
 $1.2 = 0.3t$
 $t = 4$

M1: Equation involving both $\mathbf{i}$ and $\mathbf{j}$ components of their velocity. Could have incorrect signs, for example $2.2 + 0.9t = -(1 + 1.2t)$.

M1

A1F: Correct equation.

A1F

A1F: Correct time, for their acceleration.

A1F

3

[11]

Q17.

(a) Using $\mathbf{F} = m\mathbf{a}$,

$$400 \cos \frac{\pi}{2} t \mathbf{i} + 600t^2 \mathbf{j} = 200 \mathbf{a}$$

M1

$$\mathbf{a} = 2 \cos \frac{\pi}{2} t \mathbf{i} + 3t^2 \mathbf{j}$$

A1

2

(b) $\mathbf{v} = \int \mathbf{a} \, dt$

M1 for either $\int \mathbf{a} \, dt$ or 1 of 2 terms correct

M1

$$= \frac{4}{\pi} \sin \frac{\pi}{2} t \mathbf{i} + t^3 \mathbf{j} + \mathbf{c}$$

m1 for + $\mathbf{c}$

A1m1

When $t = 4$, $\mathbf{r} = -3\mathbf{i} + 56\mathbf{j}$,
 $64\mathbf{j} + \mathbf{c} = -3\mathbf{i} + 56\mathbf{j}$

m1

$$\therefore \mathbf{c} = -3\mathbf{i} - 8\mathbf{j}$$

$$\therefore \mathbf{v} = \left(\frac{4}{\pi} \sin \frac{\pi}{2} t - 3\right)\mathbf{i} + (t^3 - 8)\mathbf{j}$$

$\frac{2}{\pi}$
 Do not accept $\frac{2}{\pi}$ Accept 1.27 for $\frac{4}{\pi}$

A1

5

- (c) When particle is moving due west, northerly component is zero

M1

$$\therefore t^3 - 8 = 0$$

A1ft

$$t = 2$$

A1

3

- (d) When $t = 2$, $\mathbf{v} = -3\mathbf{i} + 0\mathbf{j}$

B1ft

Speed of particle is 3 m s^{-1}

B1 for change -3 to $+3$

B1

2

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

[12]

Q18.

(a) $0^2 = (30 \sin 35^\circ)^2 + 2 \times (-9.8)s$

M1: Equation to find the max height, with $v = 0$

M1

A1: Correct equation

A1

M1: Solving for the height

M1

$$s = \frac{(30 \sin 35^\circ)^2}{2 \times 9.8} = 15.1 \text{ m}$$

A1: Correct height

A1

4

(b) $28^2 = (30 \cos 35^\circ)^2 + v_y^2$

M1: Equation to find vertical component

M1

A1: Correct equation

A1

$$v_y = \sqrt{28^2 - (30 \cos 35^\circ)^2}$$

M1: Solving equation

M1

$= 13.4198 = 13.4 \text{ m s}^{-1}$ **AG**

A1: Correct speed from correct working.

A1

4

(c) $\tan \theta = \frac{13.4}{30 \cos 35^\circ}$

M1: Expression to find the angle.

M1

$\theta = 28.6^\circ$

A1: Correct angle.

A1

2

[10]

Q19.

(a) Resultant $= (6\mathbf{i} - 3\mathbf{j}) + (3\mathbf{i} + 15\mathbf{j})$
 $= 9\mathbf{i} + 12\mathbf{j}$

M1: Summing the two vectors

M1

A1: Correct resultant

A1

2

(b) Magnitude $= \sqrt{9^2 + 12^2}$

$= 15 \text{ N}$

M1: Finding magnitude with an addition sign.

M1

A1F: Correct magnitude based on their answer to part (a).

A1F

2

(c) $1.5m = 9$ $2m = 12$

or

$m = 6 \text{ kg}$ $m = 6 \text{ kg}$

M1: Applying Newton's second law to one or both of the components.

M1

A1F: Correct mass, follow through their answer to part (a). Do not award this mark if vector division with 2 components has been used, eg

$$\frac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}} = 6$$
or $6\mathbf{i} + 6\mathbf{j}$ etc without a correct previous statement gives M0A0

A1F

2

(d) (i) $\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j})t^2$

M1: Using a constant acceleration equation to find the position vector with $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$

M1

A1: Correct position vector.

A1

2

(ii) $\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j}) \times 2^2 = 3\mathbf{i} + 4\mathbf{j}$

$d = \sqrt{(3)^2 + (4)^2}$

M1: Finding the position vector when $t = 2$

($\mathbf{r} = (1.5\mathbf{i} + 2\mathbf{j}) \times 2 = 3\mathbf{i} + 4\mathbf{j}$ scores M0 unless it is clear how the 2 was obtained, possibly by a correct formula in (d) (i))

M1

$$= \sqrt{25} = 5$$

A1: Correct distance

A1

2

[10]

Q20.

(a) $\mathbf{v} = \frac{dr}{dt}$

M1

$$\mathbf{v} = (e^{\frac{1}{2}t} - 8)\mathbf{i} + (2t - 6)\mathbf{j}$$

i terms

A1

j terms

A1

3

(b) (i) When $t = 3$, $\mathbf{v} = -3.52\mathbf{i}$

Accept $(e^{\frac{3}{2}} - 8)\mathbf{i}$

B1

Speed is 3.52 m s^{-1}

3.5 does not give 2nd B mark

B1

2

(ii) West

B1

1

(c) $\mathbf{a} = \frac{1}{2}e^{\frac{1}{2}t}\mathbf{i} + 2\mathbf{j}$

M1A1

When $t = 3$, $\mathbf{a} = \frac{1}{2}e^{\frac{3}{2}} \mathbf{i} + 2\mathbf{j}$ or $2.24\mathbf{i} + 2\mathbf{j}$

A1

3

(d) Using $\mathbf{F} = m\mathbf{a}$:

Accept $\mathbf{F} = 7\mathbf{a}$

M1

$$\mathbf{F} = 7 \left(\frac{1}{2}e^{\frac{3}{2}}\mathbf{i} + 2\mathbf{j} \right)$$

$\therefore$ Magnitude of force is $7 \left(\left(\frac{1}{2}e^{\frac{3}{2}} \right)^2 + 2^2 \right)^{\frac{1}{2}}$

M1

$$\mathbf{F} = 21.025$$

$$\mathbf{F} = 21.0$$

Accept 21

A1

3

[12]

Q21.

(a) $\mathbf{F} = 5\mathbf{j} + 8\mathbf{i} - 7\mathbf{j} = 8\mathbf{i} - 2\mathbf{j}$

Adding the two forces. For incorrect answers, evidence of adding must be seen

M1

Correct resultant

A1

2

(b) $F = \sqrt{8^2 + 2^2} = \sqrt{68} = 8.25 \text{ N}$

Finding magnitude (must see addition and not subtraction)

M1

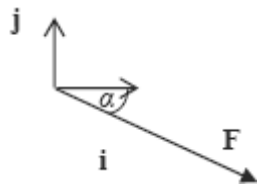
Correct magnitude

Accept $2\sqrt{17}$, $\sqrt{68}$ or AWRT 8.25 (eg 8.246)

A1F

2

(c)



*Diagram with force in the correct quadrant
and with correct direction shown by an arrow.*

B1

$$\tan \alpha = \frac{2}{8}$$

*Using trig to find angle: if tan, 8 in
denominator; if sin or cos, 8.25 or their
answer to part (b) in denominator*

M1

$$\alpha = 14.0^\circ$$

*Correct angle
Accept 14.1 or 14 or AWRT 14.0 (eg 14.04)
M1 and A1 not dependent on B1*

A1

3

[7]

Q22.

EXAM PAPERS PRACTICE

*Use of column vectors is acceptable
throughout this question.*

(a) $\mathbf{v} = 20\mathbf{i} + (-0.4\mathbf{i} + 0.5\mathbf{j})t$

*Use of constant acceleration equation to
find expression for $\mathbf{v}$*

M1

Any correct expression.

A1

2

(b) $\mathbf{v} = (20 - 0.4t)\mathbf{i} + 0.5t\mathbf{j}$

*Simplifying $\mathbf{v}$. (May be implied.)
(Missing brackets may be condoned if
followed by correct working.)*

M1

$$20 - 0.4t = 0$$

Putting i component equal to zero

m1

$$t = \frac{20}{0.4} = 50 \text{ seconds}$$

Correct time

Candidates who are able to see the correct time without supporting working gain full marks.

Condone $\frac{20\mathbf{i}}{0.4\mathbf{i}} = 50$

A1

3

(c) $\mathbf{r} = 20\mathbf{i} \times t + \frac{1}{2} (-0.4\mathbf{i} + 0.5\mathbf{j}) \times t^2$

Use of constant acceleration equation to find expression for r

Any correct expression

M1

A1

2

(d) (i) $\mathbf{r} = 20\mathbf{i} \times 100 + \frac{1}{2} (-0.4\mathbf{i} + 0.5\mathbf{j}) \times 100^2$

Substituting $t = 100$ into their expression for r (dependent on M1 in part (c))

$$= 2000\mathbf{i} - 2000\mathbf{i} + 2500\mathbf{j}$$

$$= 2500\mathbf{j}$$

Correct simplified position vector ie 2500j

m1

A1

Therefore due north

Conclusion that helicopter is due north provided their position vector is of the form $k\mathbf{j}$, where $k > 0$

Note if integration is used there is no need to prove that the constant is zero.

Note marks for (d) (i) can be awarded if part (c) scores zero.

A1

(ii) $\mathbf{v} = (20 - 0.4 \times 100)\mathbf{i} + 0.5 \times 100\mathbf{j}$

Substituting $t = 100$ into their expression for $\mathbf{v}$ (dependent on M1 in part (a)) or use of other constant acceleration equation and their position vector (dependent on M1 in part (c))

m1

$= -20\mathbf{i} + 50\mathbf{j}$

Correct simplified velocity

A1

$v = \sqrt{20^2 + 50^2} = 53.9$

Correct speed (Accept $10\sqrt{29}$)

A1

Note marks for (d) (ii) can be awarded if part (a) scores zero.

3

[13]

Q23.

(a) $\mathbf{v} = 1.2\mathbf{i} + (-0.9 \sin t)\mathbf{j}$

Convincingly differentiation

M1A1

EXAM PAPERS PRACTICE

(b) (i) Speed = $\sqrt{1.2^2 + 0.9^2 \sin^2 t}$

M1A1

2

(ii) Maximum speed = $\sqrt{1.2^2 + 0.9^2}$

Use of $\sin t = 1$

m1

$= 1.5$

A1

2

[6]

Q24.

(a) $\mathbf{v} = 4\mathbf{i} + (-3\mathbf{i} + 12\mathbf{j})t$

M1

use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

A1

2

(b) $t = 0.5, \mathbf{v} = 2.5\mathbf{i} + 6\mathbf{j}$

Ft 2 terms and t subs

B1ft

Speed = $\sqrt{(2.5)^2 + 6^2}$

2 terms

M1

Speed = 6.5 ms^{-1}

ft 2 terms

A1ft

3

[5]

Q25.

(a) $\mathbf{v} = (6t^2 - 2t)\mathbf{i} + (1 - 12t^2)\mathbf{j}$

differentiating both components

M1

one component correct

A1

second component correct

A1

3

(b) (i) $\mathbf{v}\left(\frac{1}{3}\right) = \left(\frac{6}{9} - \frac{2}{3}\right)\mathbf{i} + \left(1 - \frac{12}{9}\right)\mathbf{j} = -\frac{1}{3}\mathbf{j}$

substituting the value for t into their v

M1

correct velocity

A1

2

(ii) Travelling due south

correct description (Follow through from $\mathbf{v} = \pm k\mathbf{j}$)

A1ft

1

(c) $\mathbf{a} = (12t - 2)\mathbf{i} - 24t\mathbf{j}$

differentiating their velocity

M1

$\mathbf{a}(4) = 46\mathbf{i} - 96\mathbf{j}$

correct acceleration at time t

A1

correct acceleration at $t = 4$

A1

3

(d) $\mathbf{F} = 6(46\mathbf{i} - 96\mathbf{j}) = 276\mathbf{i} - 576\mathbf{j}$

apply Newton's second law correctly

M1

$F = \sqrt{276^2 + 576^2} = 639 \text{ N}$

finding magnitude

M1

correct magnitude

A1

3

or

$a = \sqrt{46^2 + 96^2} = 106.45$

$F = 6 \times 106.45 = 639 \text{ N}$

[12]

Q26.

(a) $-\mathbf{i} + \mathbf{j} = 2\mathbf{i} - \mathbf{j} + 10\mathbf{a}$

Use of velocity equation

M1

$\mathbf{a} = -0.3\mathbf{i} + 0.2\mathbf{j}$

Correct equation

A1

Correct $\mathbf{a}$

A1

3

(b) $\mathbf{r} = (2\mathbf{i} - \mathbf{j})t + \frac{1}{2}(-0.3\mathbf{i} + 0.2\mathbf{j})t^2 + 20\mathbf{i}$

Use of constant acceleration equation for position

M1

$= (2t - 0.15t^2 + 20)\mathbf{i} + (-t + 0.1t^2)\mathbf{j}$

Correct i component

A1

Correct j component

A1ft

3

ft incorrect acceleration

(c) (i) $\mathbf{r}(20) = (2 \times 20 - 0.15 \times 20^2 + 20)\mathbf{i} + (-20 + 0.1 \times 20^2)\mathbf{j}$
 $= 0\mathbf{i} + 20\mathbf{j}$

Substituting $t = 20$ into their expression for $\mathbf{r}$

M1

so due north of origin

Correct conclusion from correct working

A1

2

(c) (ii) $\mathbf{v}(20) = 2\mathbf{i} - \mathbf{j} + 20(-0.3\mathbf{i} + 0.2\mathbf{j}) = -4\mathbf{i} + 3\mathbf{j}$

$v(20) = \sqrt{4^2 + 3^2} = 5 \text{ ms}^{-1}$

Finding velocity at $t = 20$

M1

Correct velocity

A1

Finding magnitude

m1

Correct speed

A1ft

4

ft incorrect acceleration

[12]

Q27.

(a) $\mathbf{v} = 4\mathbf{j} + (3\mathbf{i} - 5\mathbf{j})t$

M1: Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$ and

$\mathbf{u} \neq 0$ or Integration

A1: Correct expression

M1A1

2

(b) $\mathbf{v} = 3t\mathbf{i} + (4 - 5t)\mathbf{j}$

M1: $\mathbf{j}$ component of velocity equal to zero

M1

$$4 - 5t = 0$$

$$t = 0.8 \text{ s}$$

A1: Correct t

A1

2

(c) $\mathbf{v} = 12\mathbf{i} - 16\mathbf{j}$

M1: Finding velocity when $t = 4$

M1

$$v = \sqrt{12^2 + 16^2} = 20 \quad \text{AG}$$

A1: Correct velocity

A1

dM1: Finding the magnitude

dM1

A1: Correct speed from correct working

A1

4

[8]

Q28.

(a) $\mathbf{a} = 8 \cos t \mathbf{i} - 4 \sin t \mathbf{j}$

M1 differentiation

M1A1

$$\mathbf{F} = 2 \cos t \mathbf{i} - \sin t \mathbf{j}$$

A1

3

(b) (i) $|\mathbf{F}| = \sqrt{(4 \cos^2 t + \sin^2 t)}$

Magnitude

M1

$$= \sqrt{(3 \cos^2 t + 1)}$$

m1

CAO

A1

3

(ii) $|\mathbf{F}| \leq 4$

B1 each value (at end of range)

B1B1

2

[8]

Q29.

Answers in (a), (b) and (c)(i) must be vectors

(a) $\mathbf{a} = 4\mathbf{i} + 3\mathbf{j}$

Must be vector, both i and j present

B1

1

(b) $\mathbf{v} = 2t^2\mathbf{i} + 3t\mathbf{j}$

Integration of acceleration attempted

M1

ft a, both i and j present

A1F

2

(c) (i) $\mathbf{r} = \int_0^t (2t^2\mathbf{i} + 3t\mathbf{j}) dt$

Integration attempted

M1

$$= \frac{2t^3}{3}\mathbf{i} + \frac{3t^2}{2}\mathbf{j}$$

A1 each term. Condone i and j missing.

–1 for constants present,

–1 for both terms unsimplified

A1A1

3

$$(ii) \left. \begin{array}{l} \mathbf{v} \text{ parallel to } \mathbf{i} + \mathbf{j} \\ 2t^2 = 3t \end{array} \right\}$$

M1

$$t = \frac{3}{2}$$

Solve, ft $\mathbf{v}$ with vector components

A1F

$$\mathbf{r} = \frac{9}{4}\mathbf{i} + \frac{27}{8}\mathbf{j}$$

A1F

$$\text{Distance} = \sqrt{\left(\frac{9}{4}\right)^2 + \left(\frac{27}{8}\right)^2}$$

Magnitude of $\mathbf{r}$ in any form

M1

$$= 4.06 \text{ (4.056)}$$

Numerical answer, ft candidate's $\mathbf{r}$ and t

A1F

5

[11]

Q30.

EXAM PAPERS PRACTICE

$$(a) \quad \mathbf{F} = \begin{pmatrix} 6 \\ -2.5 \end{pmatrix} \text{ N}$$

B1 each component

B1 B1

2

$$(b) \quad |\mathbf{F}| = \sqrt{6^2 + (-2.5)^2}$$

Must see +

M1

$$= 6.5 \text{ N}$$

ft from vector in (a)

A1F

2

Q31.

(a) $x = 7t$

B1

$$y = 0 + \frac{1}{2}gt^2$$

Accept ±

M1

$$y = 4.9t^2$$

Accept ±

A1

3

(b) $t = \frac{x}{7}$

Attempt at substitution, or use of equation of trajectory with $V = 7$ & $\alpha = 0$

M1

$$y = 4.9 \left(\frac{x}{7} \right)^2$$

$$y = \frac{x^2}{10}$$

CAO

A1

2

(c) $8.1 = \frac{x^2}{10}$

Full method for x, accept ±

M1

$$x = 9\text{m}$$

AWRT 9.0 if two stages used

A1

2

(d) vert: $v^2 = 0 + 2 \times 9.8 \times 8.1$

For M1 Accept $\pm$,
 for A1 consistency of signs needed

M1A1

$$u = 7$$

B1

$$\text{speed}^2 = (7^2 + 12.6^2)$$

M1

$$\text{speed} = 14.4 \text{ ms}^{-1}$$

$$(14.414)$$

A1F

5

[12]

Q32.

(a) $\mathbf{v} = 6\mathbf{i} + 2t\mathbf{j}$

M1: differentiation attempted and vector quantity for $\mathbf{v}$ given

M1A1

2

(b) $\text{sp} = \sqrt{(6^2 + 4t^2)} \text{ ms}^{-1}$

M1: sum of squares attempted giving scalar expression

A1 : all correct, accept $(2t)^2$

f.t. $\mathbf{v}$ with 2 components

M1A1F

2

(c) $\sqrt{(6^2 + 4t^2)} = 6\sqrt{2}$

o.e.; scalar expression for $\mathbf{v}$ in terms of t from (b) used

M1

$$36 + 4t^2 = 36 \times 2$$

$$t = 3$$

f.t. minor slip in (b) provided t is positive solution of quadratic eqn

A1F

2

[6]

Q33.

(a) $x = 7t, y = 7t - \frac{1}{2} \times 9.8t^2$ $\begin{pmatrix} 7t \\ 7t - 4.9t^2 \end{pmatrix}$

B1 horiz

M1 use of $s = ut + \frac{1}{2}at^2$

M1A1 for vert, accept 10 or g

B1M1A1

3

(b) $t = \frac{x}{7}$

B1

$$y = 7 \times \frac{x}{7} - \frac{1}{2} \times 9.8 \times \frac{x^2}{49}$$

M1A1

$$y = x - 0.1x^2$$

A1

4

Alternative (full verification method)

$x = 7t$ used B1

subs into $y = x - 0.1x^2$

M1A1

$y = 7t - 4.9t^2$ as in (a)

A1

Alternative: (use equation of path) All rights reserved.

$$y = x \tan \alpha - \frac{gx^2}{2v^2 \cos^2 \alpha}$$

45° used

B1

subs

M1A1

simplify

A1

(c) $x = 4, y = 4 - 0.1 \times 4^2$

$$\text{or } t = \frac{4}{7} (0.571(43))$$

M1

$$y = 2.4$$

A1

above wire as $2.4 > 2$

ft slip in y provided y positive

A1F

3

(d) $y = 0$
 $x - 0.1x^2 = 0$

M1

$$x(1 - 0.1x) = 0$$

m1 A1

$$x = 10$$

A1F

4

Alternative: (use of coordinate equations)

$$y = 0, 7t - 4.9t^2 = 0$$

M1

$$\Rightarrow t = \frac{10}{7} \quad (1.43)$$

m1A1

$$x = 7t = 10$$

A1F

(Accept 10.01 if $t = 1.43$ used)

Alternative: (use of symmetry)

$$v = 0, 0 = 7 - 9.8t$$

M1

$$t = \frac{5}{9.8}$$

A1

$$x = 7 \times t \times 2$$

m1

$$x = 10 \text{ (acc 9.996)}$$

A1F

Alternative: (use of equation for Range)

$$R = \frac{2V^2 \sin \alpha \cos \alpha}{g}$$

Correct formula used M1, 45° used m1

Correct subs A1,

result A1F, AWRT 10m, ft one slip.

(e) $\text{Speed}^2 = 7^2 + 7^2$

M1

$= 9.90 \text{ ms}^{-1}$

Accept 9.9, $7\sqrt{2}$,

or 9.91 if $t = 1.43$ used in downward motion

A1

2

[16]

Q34.

(a) $\mathbf{v} = -4e^{-t} \mathbf{i} + (6 - 3e^{-t})\mathbf{j}$

Differentiating position vector

M1

$t = 0$ A1

Correct velocity

$\mathbf{v} = -4\mathbf{i} + 3\mathbf{j}$

ag Substituting $t = 0$ to obtain initial velocity

A1

3

(b) $\mathbf{a} = 4e^{-t} \mathbf{i} + 3e^{-t} \mathbf{j}$

Differentiating velocity

M1

Correct acceleration

A1

2

(c) $\mathbf{a} = 4\mathbf{i} + 3\mathbf{j}$

Finding acceleration when $t = 0$

M1

$a = \sqrt{4^2 + 3^2} = 5$

Correct magnitude

A1

2

(d) $\mathbf{v} \rightarrow 0\mathbf{i} + 6\mathbf{j}$

For i component

B1

For j component

B1

2

[9]

Q35.

(a) $19\mathbf{i} - 25\mathbf{j} = \frac{1}{2} \mathbf{a} \times 10^2 + 9\mathbf{i} + 10\mathbf{j}$

Using both position vectors to form a constant acceleration equation

M1

$50\mathbf{a} = 10\mathbf{i} - 35\mathbf{j}$

Correct equation

Solving for a

A1 M1

$\mathbf{a} = 0.2\mathbf{i} - 0.7\mathbf{j}$

Correct a

A1

4

(b) $\mathbf{v} = 10(0.2\mathbf{i} - 0.7\mathbf{j})$

use of $\mathbf{v} = \mathbf{at}$

M1

$= 2\mathbf{i} - 7\mathbf{j}$

Correct velocity

A1

$v = \sqrt{2^2 + 7^2} = 7.28 \text{ ms}^{-1}$

Finding speed from velocity

Correct speed

m1 A1

4

(c) $15.4 = \frac{1}{2} \times 0.2 \times t^2 + 9$
Finding t from one component

M1

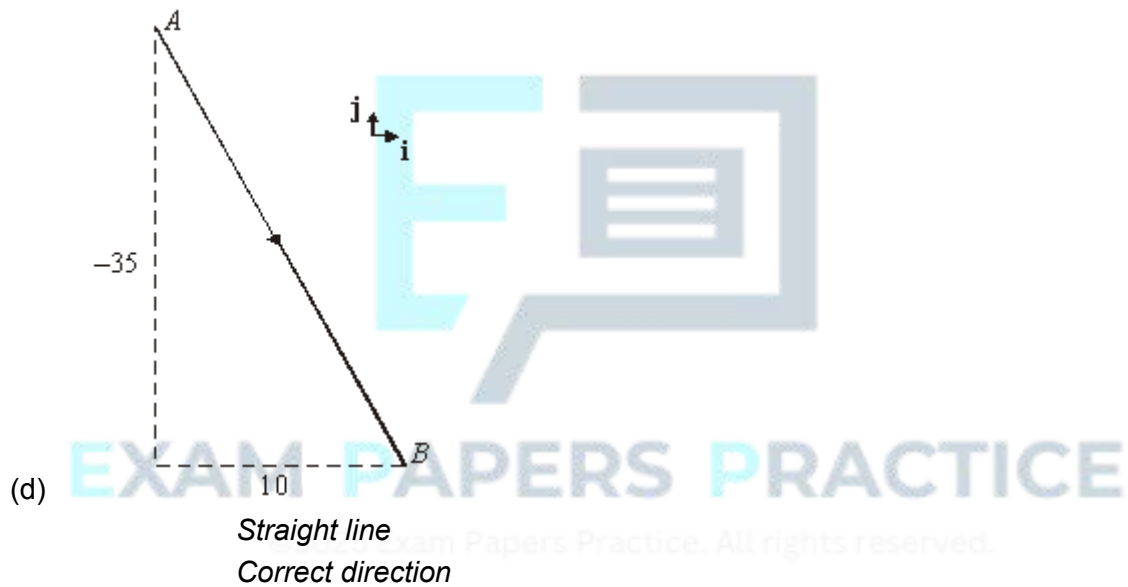
$t = 8$
Correct t

A1

$\frac{1}{2} \times (-0.7) \times 8^2 + 10 = -12.4$
Using $t = 8$ with other component
Correct result

M1 A

4



B1 B1

2

[14]